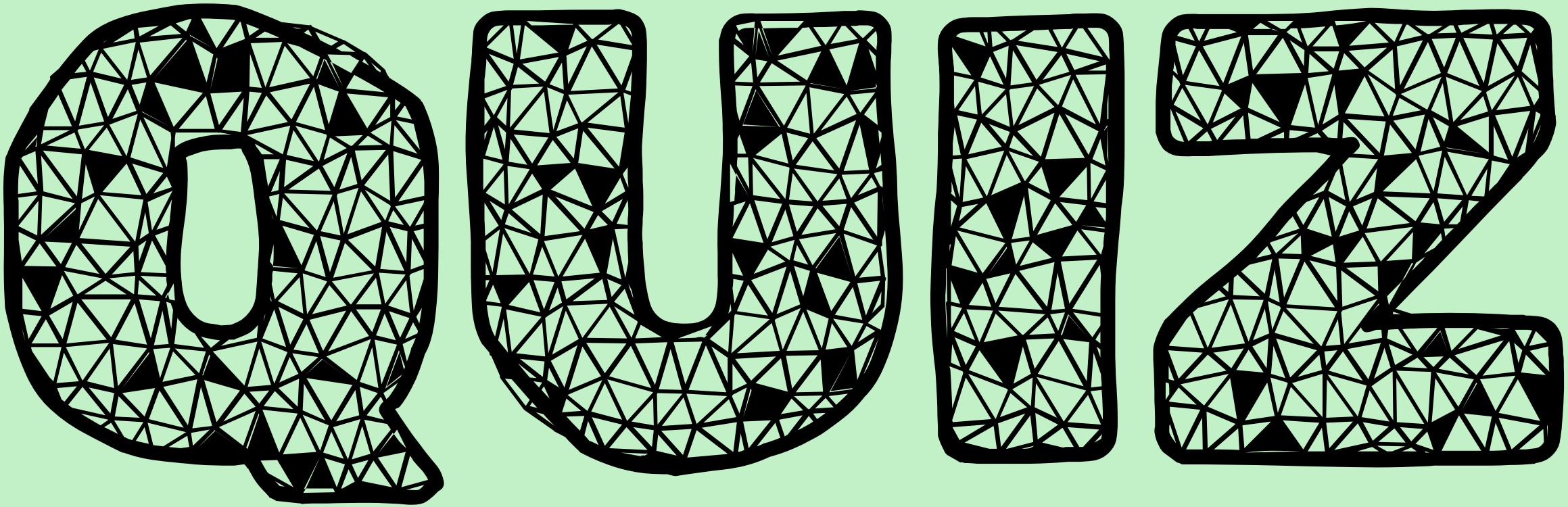
Information Assurance and Security – ACIT4630

Iman Anooshehpour
Week 10

BRITISH COLUMBIA
INSTITUTE TECHNOLOGY

BCIT™

PWD: 0-trust



Learning Objectives

- ❑ Explain the problem addressed by Network Address Translation (NAT)
- ❑ Explain functionality and use cases of different kinds of a firewall
- ❑ Compare a zero-trust architecture with a traditional security architecture.

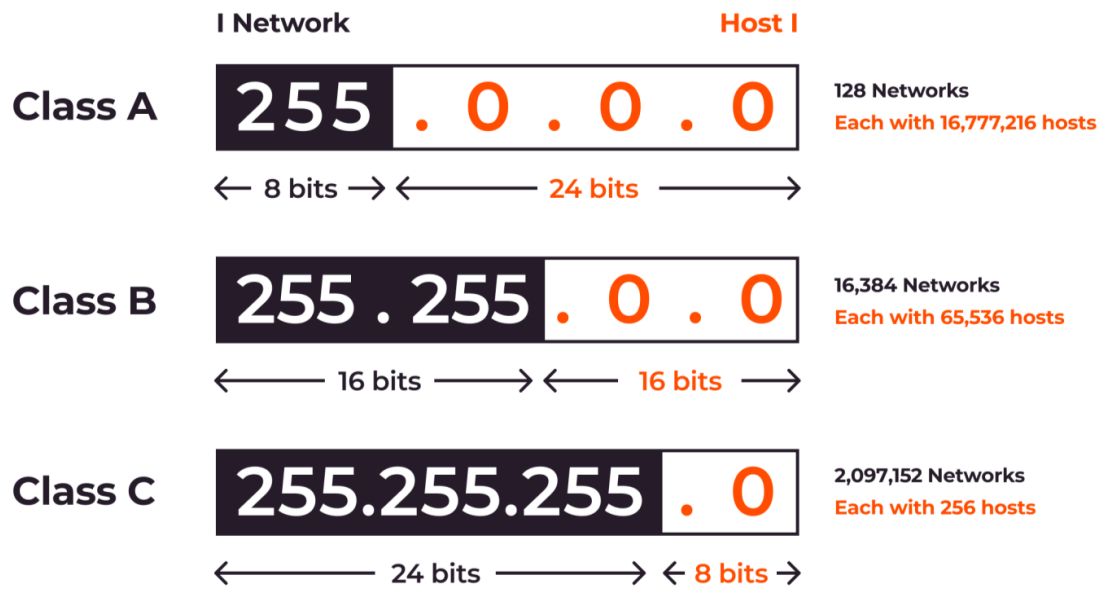
IPv4 vs IPv6

IPv4	vs.	IPv6
Deployed 1981		Deployed 1998
32-bit IP address		128-bit IP address
4.3 billion addresses Addresses must be reused and masked ?		7.9×10^{28} addresses Every device can have a unique address
Numeric dot-decimal notation 192.168.5.18		Alphanumeric hexadecimal notation 50b2:6400:0000:0000:6c3a:b17d:0000:10a9 (Simplified - 50b2:6400::6c3a:b17d:0:10a9)
DHCP or manual configuration		Supports autoconfiguration

Google IPv6 Adoption Statistics

Subnetting

A **subnetwork** or **subnet** is a logical subdivision of an IP network. The practice of dividing a network into two or more networks is called **subnetting**.



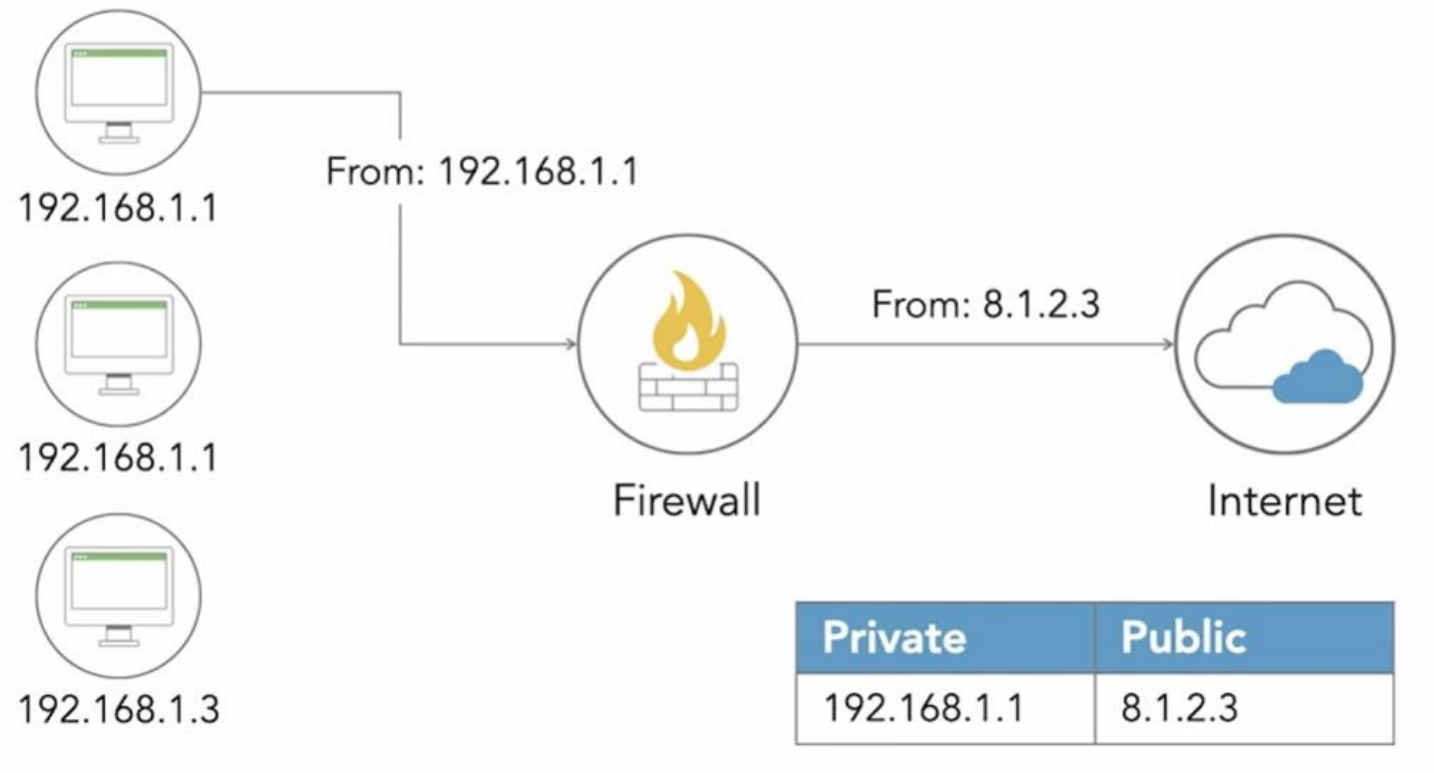
Fixed Bits	Net Bits	Host Bits	Networks (2^n)	Hosts ($2^h - 2$)
0 (1 bit)	7	24	2^7 = 128	16,777,214
10 (2 bits)	14	16	2^{14} = 16,384	65,534
110 (3 bits)	21	8	2^{21}= 2,097,152	254

BCIT's Class B Subnet 😎

NETBLOCK	COMPANY	NUM OF IPS
142.232.0.0/16	British Columbia Institute of Technology	65,536

Source: https://ipinfo.io/AS4476

Network Address Translation

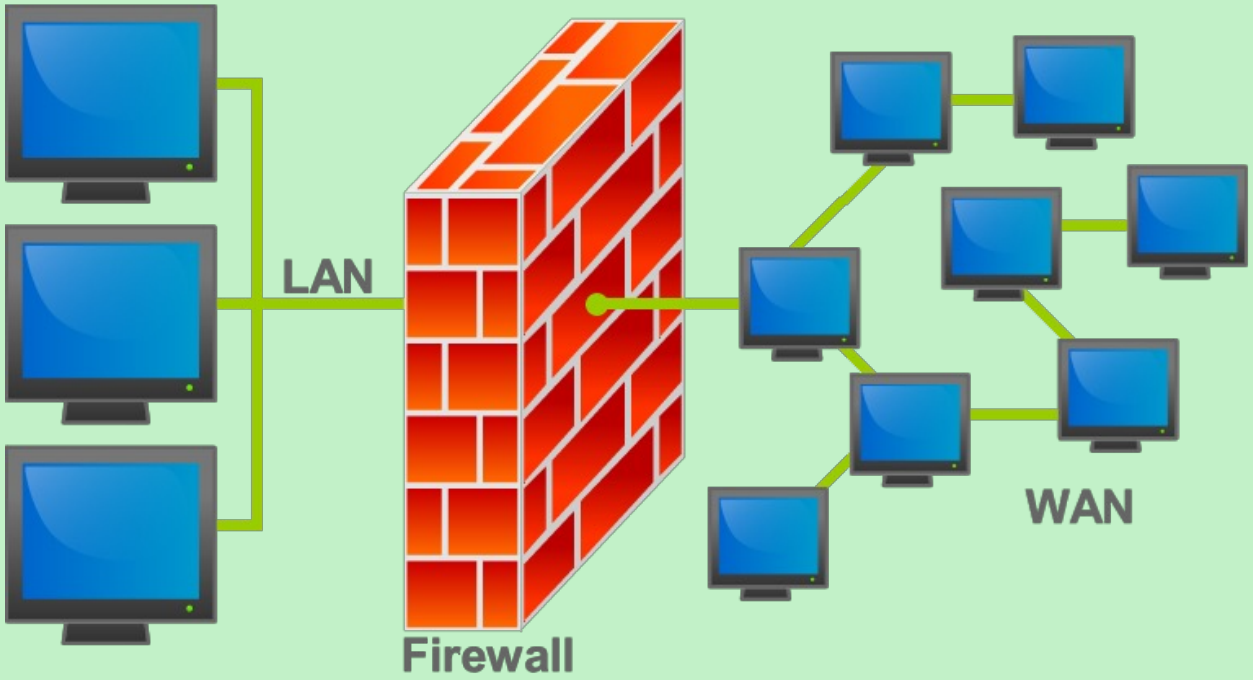


What are some security concerns with NAT?

Security Concerns with NAT

- End-to-End Traceability
- Port Forwarding Vulnerabilities
- IP Masquerading Risks
- Security Policy and Tool Effectiveness
- Difficulty in Monitoring and Logging
- NAT Table Overflow
- Dependence on Specific IP Addresses

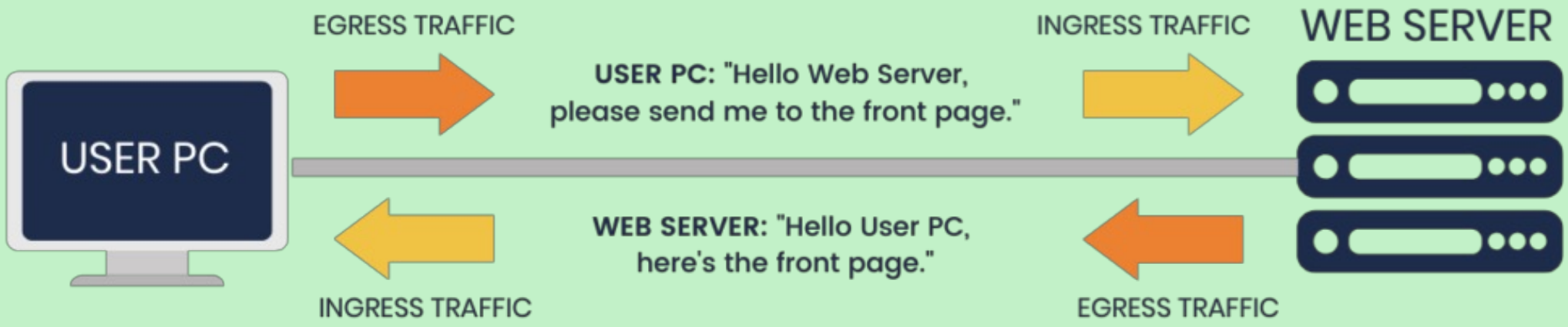
Firewalls on the Network



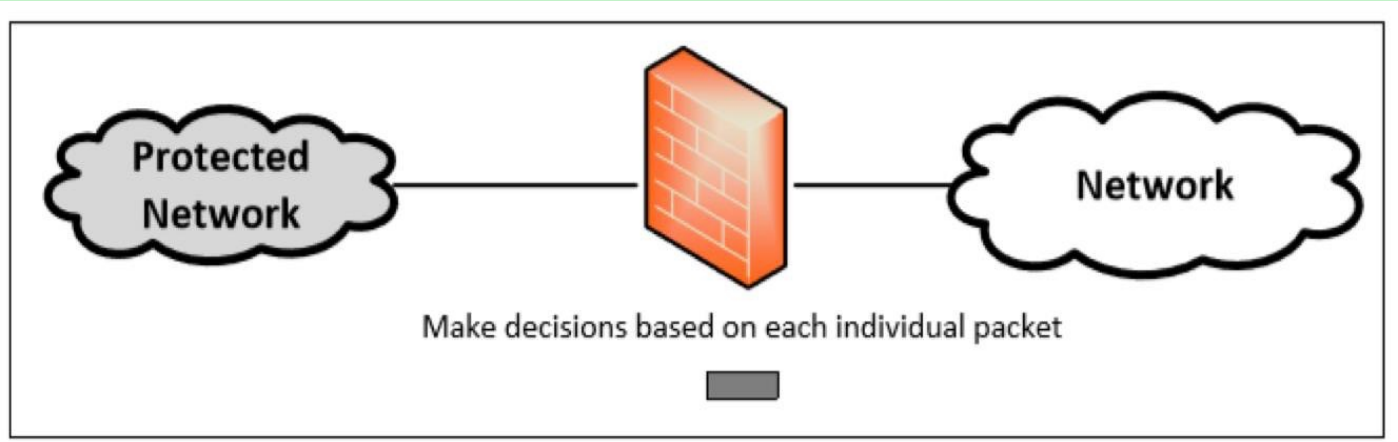
What are different actions the firewall can do when receiving a packet?

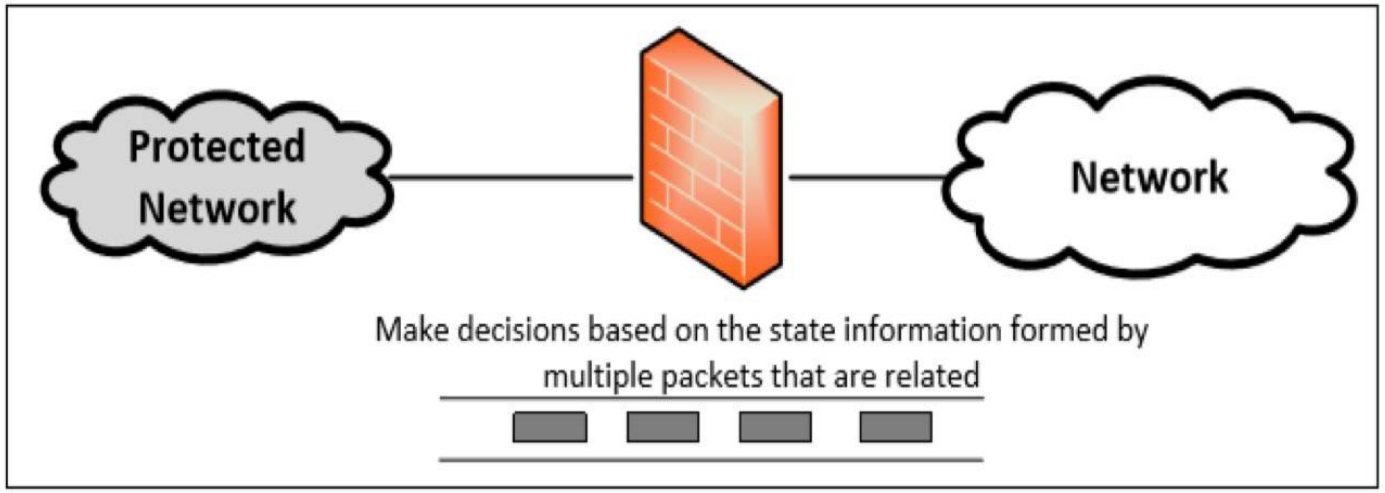
Ingress vs Egress filtering

- How are ingress filtering different from egress filtering?
- What are some use-cases for egress filtering?

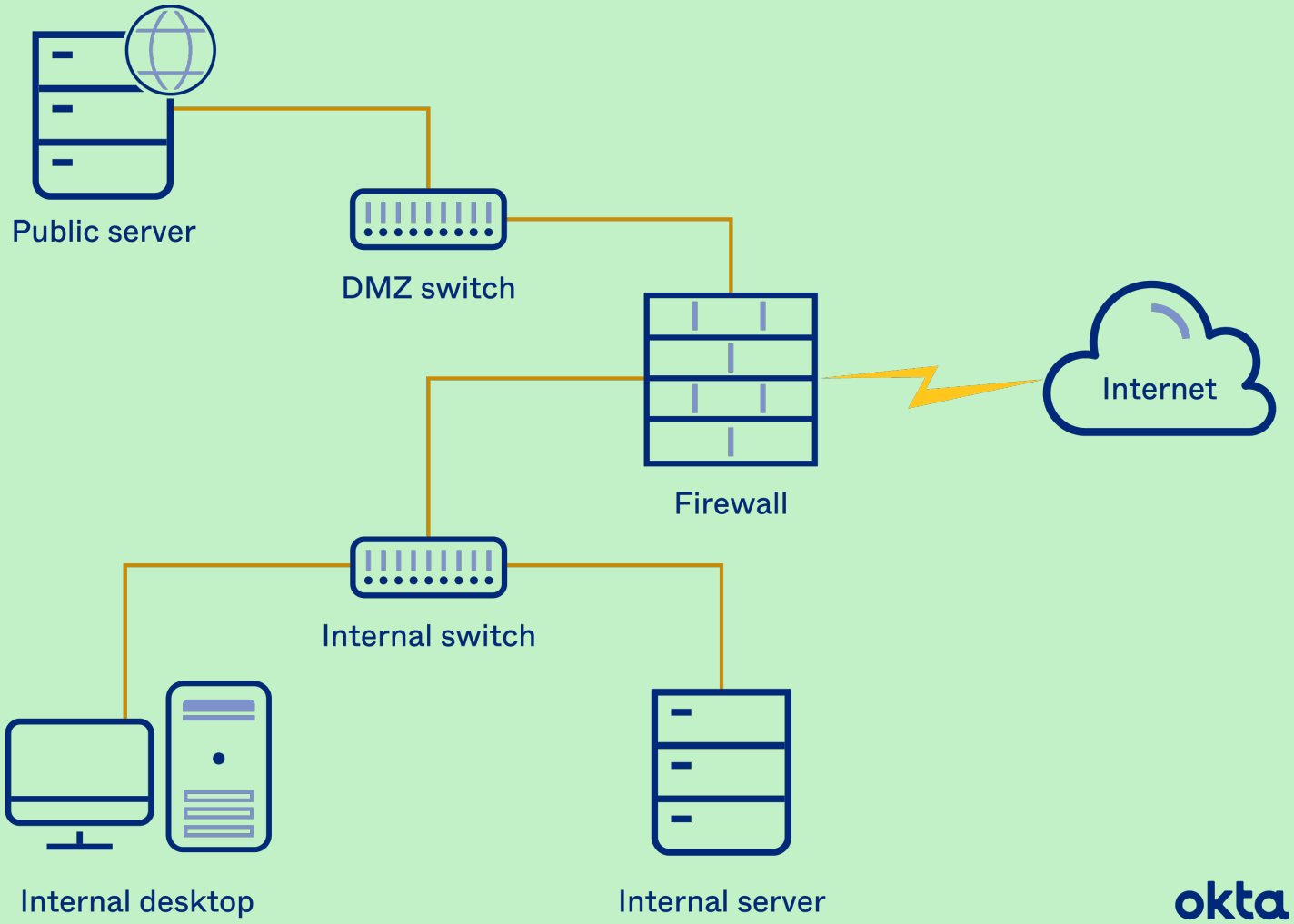


Stateless vs Stateful filtering





Demilitarized Zone Network



Zero-Trust Architecture

- How is a zero-trust network architecture different from a traditional (castle-and moat) security architecture?
- How is the level of trust for each user granted in a zero-trust network?
- SIEM aids in monitoring user activities and traffic, facilitating continuous verification and anomaly detection
- SOAR can be used to automatically respond to security alerts, implement security policies, and manage access controls

Zero-Trust vs Castle-and-Moat

- Core Concept:
  - Zero-Trust: Trust no one, verify everything, regardless of location.
  - Castle-and-Moat: Strong perimeter defense; once inside, users are trusted.
- Access Control:
  - Zero-Trust: Access is granted on a need-to-know basis, with constant verification.
  - Castle-and-Moat: Open access within the network once inside the perimeter.
- Attack Surface:
  - Zero-Trust: Reduces risk by limiting access at each point.
  - Castle-and-Moat: Vulnerable if perimeter is breached.
- Adaptability:
  - Zero-Trust: Better suited for modern, distributed, and cloud-based environments.
  - Castle-and-Moat: More traditional, suited to localized, single-network setups.

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LAB TIME

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